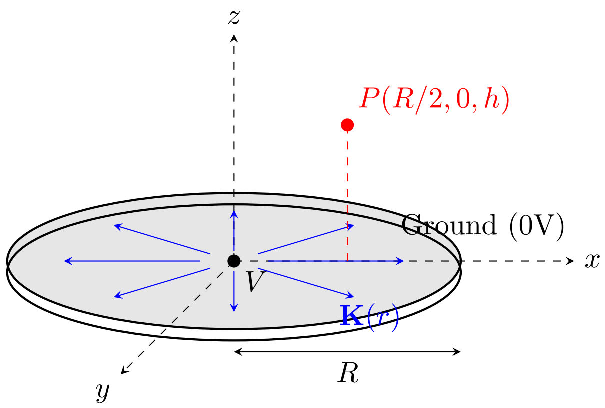
1. A thin conducting disk of radius R, thickness t ($t \ll R$), and resistivity ρ is placed in the xy-plane. The center of the disk is maintained at a potential V, while the outer circumference ($r = R$) is grounded (0V). Assume current flows uniformly and radially from the center to the edge.

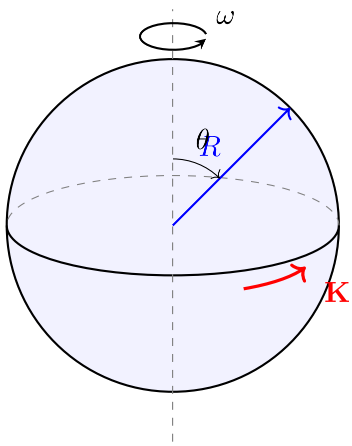
   (a) Determine the surface current density $\mathbf{K}(r)$ as a function of the distance r from the center.

   (b) Set up the integral for the magnetic field $\mathbf{B}$ at a point $P = (\frac{R}{2}, 0, h)$, where h is the height above the disk.

   (c) Determine which components (B_x, B_y, B_z) are non-zero at point P.



2. A thin, non-conducting small spherical shell of radius $R << r$ carries a uniform surface charge density σ. The shell rotates about the z-axis with a constant angular velocity $\omega = \omega\hat{k}$.



- Determine the expression for the surface current density $\mathbf{K}(\theta)$ at any point on the shell.
- Calculate the magnetic vector potential $\mathbf{A}(\mathbf{r})$ in all space using the integral:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{\mathbf{K}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} da'$$

Show that for $r > R$, it is proportional to $\frac{\sin\theta}{r^2}\hat{\phi}$.

- Also solve this using the technique mentioned in class ie

$$d\vec{A} = \frac{\mu_0}{4\pi}\frac{d\vec{m} \times \hat{r}}{r^2}$$

- Using the relation $\mathbf{B} = \nabla \times \mathbf{A}$, derive the magnetic field $\mathbf{B}$ outside the sphere.

3. Consider a long, straight coaxial cable of infinite length. The cable consists of:

   - A solid inner conductor of radius a, carrying a total current I directed into the page.
   - A thin outer cylindrical shell of radius b, carrying an equal total current I directed out of the page.

   Assume the current density J in the inner conductor is uniform. Determine the magnetic field magnitude $B(r)$ as a function of the radial distance r from the central axis for the following regions:

   (a) $r < a$ (Inside the inner conductor)

   (b) $a < r < b$ (In the air gap)

   (c) $r > b$ (Outside the cable)

   Now, suppose the current density in the inner conductor is not uniform, but instead varies with the distance from the axis according to the relation:

$$J(r) = Cr$$

   where C is a constant. Find the magnetic field $B(r)$ for $r < a$ in terms of the total current I and the geometry of the cable.

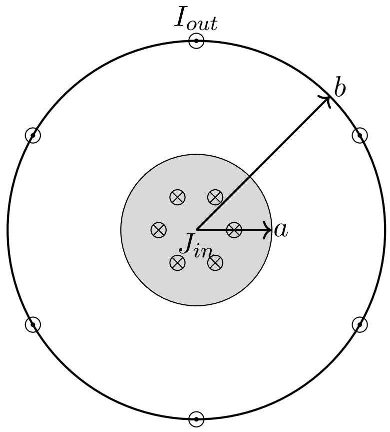


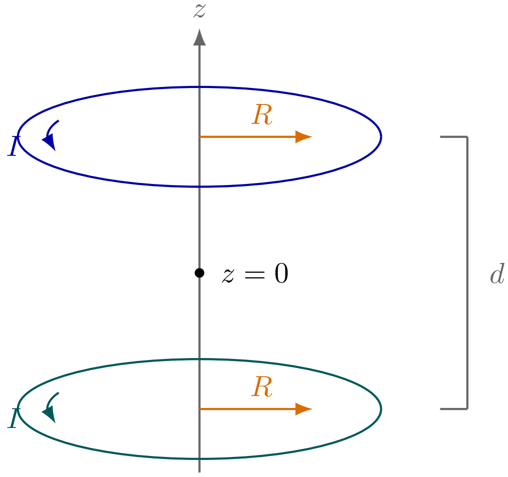
Figure 1: Cross-section of the coaxial cable.

4. Consider an infinitely long solenoid of radius R with n turns per unit length, carrying a steady current I. The axis of the solenoid coincides with the z-axis. Find the magnetic vector potential A in all space (r < R and r > R). consider $\nabla \cdot A = 0$

5. The magnetic field on the axis of a circular current loop is given by:

$$B(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

which is not uniform and falls off with increasing z. A more nearly uniform field can be produced by using two identical loops separated by a distance d along the axis.

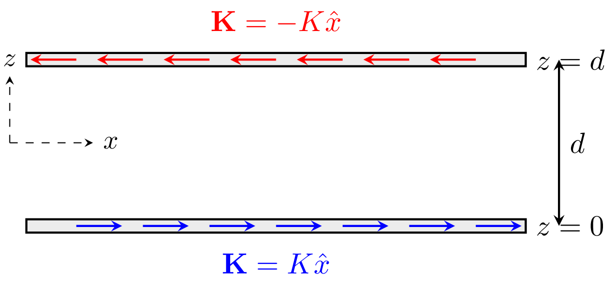


(a) Find the magnetic field $B(z)$ along the axis due to the two loops, and show that $\frac{dB}{dz} = 0$ at the midpoint ($z = 0$).

(b) Determine the separation d such that $\frac{d^2B}{dz^2} = 0$ at the midpoint. This configuration is known as a Helmholtz coil. Also find the magnetic field at the center.

(c) Plot the magnetic field $B(z)$ along the axis for both $d = R$ (Helmholtz condition) and a non-optimal separation. Comment on the uniformity of the field near the center.

6. An infinite plane at $z = 0$ carries a uniform surface current density $\mathbf{K} = K\,\hat{x}$, where K is a constant (in A/m). A second infinite plane at $z = d$ carries a surface current $\mathbf{K} = -K\,\hat{x}$ (equal magnitude, opposite direction).



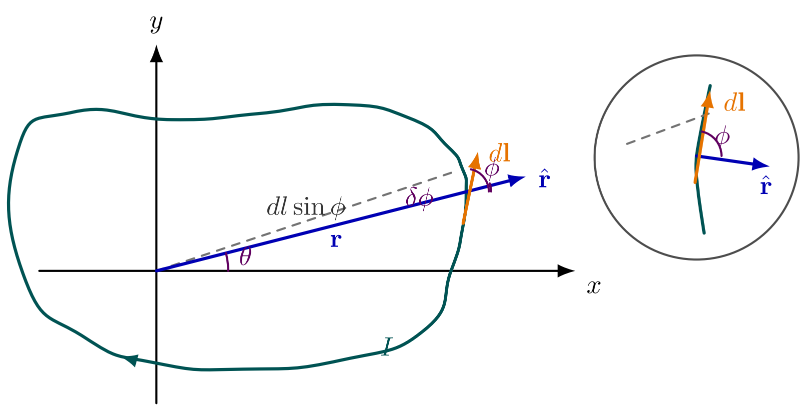
(a) Using Ampère's law with a rectangular loop, find the magnetic field $\mathbf{B}$ in each of the three regions: $z < 0$, $0 < z < d$, and $z > d$. Express your answer in terms of K, μ_0, and $\hat{y}$.

(b) This configuration is the magnetic analogue of a parallel-plate capacitor. Identify which region has a uniform confined field and which regions have zero field, and draw a clear parallel with the electric field of two oppositely charged infinite planes.

(c) The two sheets are now replaced by a single slab of finite thickness δ ($\delta \ll d$) carrying a uniform volume current density $\mathbf{J} = J_0\,\hat{x}$. Show that in the limit $\delta \to 0$ with $J_0\,\delta = K$ held fixed, you recover the surface current result from part (a).

(d) Find the magnetic vector potential $\mathbf{A}$ in all three regions for the two-sheet configuration, choosing the gauge $\nabla \cdot \mathbf{A} = 0$ and requiring $\mathbf{A}$ to be continuous at the sheets. Verify that $\mathbf{B} = \nabla \times \mathbf{A}$ reproduces part (a).

7. Consider a plane loop of wire that carries a steady current I; we want to calculate the magnetic field at a point in the plane. We might as well take that point to be the origin (it could be inside or outside the loop). The shape of the wire is given, in polar coordinates, by a specified function $r(\theta)$.



(a) Show that the magnitude of the field is

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}.$$

Hint: Start with the Biot–Savart law; note that $\hat{\mathbf{z}} = -\hat{\mathbf{r}}$, and $d\mathbf{l} \times \hat{\mathbf{r}}$ points perpendicular to the plane; show that $|d\mathbf{l} \times \hat{\mathbf{r}}| = dl \sin\phi = r\, d\theta$.

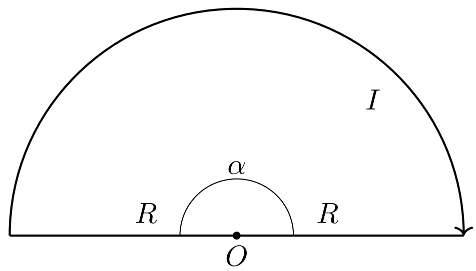
(b) Test this formula by calculating the field at the center of a circular loop.

(c) The "lituus spiral" is defined by

$$r(\theta) = \frac{a}{\sqrt{\theta}}, \quad (0 < \theta \leq 2\pi)$$

(for some constant a). Sketch this figure, and complete the loop with a straight segment along the x-axis. What is the magnetic field at the origin?

8. A wire carries a steady current I and lies in the xy-plane. It consists of two parts:



- A semicircular arc of radius R, centered at the origin, spanning from $\theta = 0$ to $\theta = \pi$.
- Two straight radial segments connecting the ends of the arc to the origin.

Thus, the wire forms a closed loop shaped like a "fan."

(a) Using symmetry and the Biot–Savart law, determine the direction of the magnetic field at the origin.

(b) Show that the contribution to the magnetic field at the origin from each radial segment is zero.

(c) Calculate the magnetic field at the origin due to the semicircular arc.

(d) Now suppose the current in the semicircular arc flows from $\theta = 0$ to $\theta = \pi$, but the current in the radial segments flows *toward* the origin instead of away from it. Does your answer for the magnetic field at the origin change? Explain carefully.

(e) Generalize your result: If instead of a semicircle, the arc subtends an angle α, what is the magnetic field at the origin?